

A Systematic Review on Ankle Injury and Ankle Sprain in Sports

Daniel Tik-Pui Fong,¹ Youlian Hong,² Lap-Ki Chan,¹ Patrick Shu-Hang Yung¹ and Kai-Ming Chan¹

- 1 Department of Orthopaedics and Traumatology, Prince of Wales Hospital, Faculty of Medicine, The Chinese University of Hong Kong, Hong Kong, China
- 2 Department of Sports Science and Physical Education, Faculty of Education, The Chinese University of Hong Kong, Hong Kong, China

Contents

Abstract	73
1. Methods	74
2. Results	76
2.1 Demographic Data	76
2.2 Prevalence of Ankle Injuries in Different Sports	77
2.3 Prevalence of Ankle Sprains in Different Sports	77
2.4 Prevalence of Ankle Injury and Ankle Sprains in Different Countries	77
2.5 Incidence of Ankle Injuries	77
2.6 Incidence of Ankle Sprains	82
3. Discussion	82
4. Conclusion	89

Abstract

This article systematically reviews epidemiological studies on sports injury from 1977 to 2005 in which ankle injury was included. A total of 227 studies reporting injury pattern in 70 sports from 38 countries were included. A total of 201 600 patients were included, with 32 509 ankle injuries. Ankle injury information was available from 14 098 patients, with 11 847 ankle sprains. Results show that the ankle was the most common injured body site in 24 of 70 included sports, especially in aeroball, wall climbing, indoor volleyball, mountaineering, netball and field events in track and field. Ankle sprain was the major ankle injury in 33 of 43 sports, especially in Australian football, field hockey, handball, orienteering, scooter and squash. In sports injuries throughout the countries studied, the ankle was the second most common injured body site after the knee, and ankle sprain was the most common type of ankle injury. The incidence of ankle injury and ankle sprain was high in court games and team sports, such as rugby, soccer, volleyball, handball and basketball. This systematic review provides a summary of the epidemiology of ankle injury in sports.

All around the world, people are participating in sports for personal interest, relaxation, health and fitness training. However, sports is one of the major causes of injuries, comparable to traffic accidents,

home and leisure accidents, occupational injuries and violence.^[1-4] Sports injuries result in pain, loss of playing or working time, as well as medical expenditure. Severe injuries may result in bone frac-

tures, functional instability, limited mobility, disability, permanent cease of sports participation, psychological problems and even death. For world-class and commercial sports teams, absence of key players due to unexpected injuries may result in defeats in major games and huge economic losses. As a result of these undesired adverse effects, scientists and medical specialists are actively working on sports injury prevention.^[5-7]

Firstly, the problem had to be identified. In 1970, Roser and Clawson^[8] conducted an epidemiological study on football injuries in young athletes. In 1971, MacIntosh et al.^[9] conducted a similar study on athletic injuries at the University of Toronto, Canada. In 1972, Garrick^[5] conducted a related study on the prevention of sports injuries, and in 1973, Garrick and Requa^[10] studied the prevention of ankle sprains. To be more specific in identifying the problem, Garrick pioneered a series of ankle-sprain epidemiology studies from 1977.^[11] Subsequent research investigated ankle injuries in different sports events including aerobic dance,^[12] children and adolescent skiing,^[13] women's gymnastics,^[14] ballet^[15] and in all sports.^[16] From 1980 onwards, numerous epidemiology studies were conducted on different sports, populations, genders, playing levels and in different countries.^[3,16-21] In general, these studies reported that the ankle is one of the most traumatised body sites in sports injuries and accounts for 10–30% of all sports injuries. In some sports, the percentage of ankle injuries is even higher (e.g. 40% in mountaineering^[22] and handball,^[23] 41% in volleyball^[24] and youth soccer,^[25] 45% in female cheerleading,^[26] 46% in artistic gymnastics,^[27] 53%^[28] and 54%^[29] in softball and 56% in women's floorball^[30]). From epidemiological data collected from 1979 to 1987, Garrick and Requa^[16] reported that the percentage of ankle injuries could be as high as 74% in softball, 76% in racquet sports and football, 77% in weight lifting and dancing, 79% in basketball and 82% in volleyball. Ankle sprain is the most common type of ankle injury and may account for >80% and even as much as 100% of ankle injury sustained in some sports (e.g. squash,^[31] orienteering,^[32] scooter,^[33] soccer,^[34] figure skating,^[35] Australian football,^[36] rugby,^[37] field hockey,^[38] handball^[39] and volleyball^[24]). In these sprains, ≈77% are lateral sprains^[40] and 73% involved rupture or tear

to the anterior talofibular ligament.^[41] In a survey conducted in 1994 on 380 athletes with 563 sprained ankles, Yeung et al.^[21] reported that the majority of these injured athletes were pursuing running and jogging activities (25%), racquet sports (20%), ball games (19%), soccer (14%) and fencing (6%). The residual problem included pain (30.2%), instability (20.4%), crepitus (18.3%), weakness (16.5%), stiffness (14.6%) and swelling (13.9%). For ankles sprained 1–4 times, the major residual problem is pain (24–28%). For ankles sprained ≥5 times, instability problems arise, which becomes the major sequelae (38%).

In 1987, van Mechelen et al.^[42] proposed a “sequence of injury prevention,” which described how sports injury-related studies come together to form the research framework. The first step is to identify the extent of the sports injury problem by epidemiology studies. The second step establishes the aetiology and mechanism of injuries and the third step designs and introduces preventive measures. Finally, the effectiveness of the preventive measures is assessed by repeating the original epidemiology study (step one). The epidemiology studies conducted by Garrick et al.^[11-16] play an important role in locating the problem of ankle sprains in sports, within the model of “sequence of injury prevention.” With this information, subsequent studies were performed to identify risk factors to injury^[43] and injury mechanism,^[44] to design prevention equipment^[45,46] and to finally evaluate the effectiveness of these preventive measures.^[47,48] The purpose of the present study is to systematically review epidemiology studies from 1977 to 2005 on sports injury, in order to identify the distribution of reported injuries among ankle and other body sites and summarise the type of ankle injury. The time span starts from the first epidemiology study on ankle sprains published by Garrick^[11] in 1977. The findings will help sports medicine specialists to identify the patterns of ankle injury in sports and to direct future research.

1. Methods

A systematic literature search of MEDLINE (from 1966), ISI Web of Knowledge (Science Citation Index Expanded, from 1985; Social Sciences Citation Index, from 1956; Arts and Humanities

Citation Index, from 1975), PubMed (from 1950); and Sport Discuss (from 1975) was conducted during the last week of January in 2006. The search keyword string was 'ankle AND (injury OR injuries OR sprain) AND (sport OR sports) AND (epidemiology OR epidemiologic OR epidemiological OR survey OR statistics OR patterns OR pattern)', which appeared in the title, abstract or keyword fields. The initial total number of identified articles from these databases was 744. After removing duplicates, the count was reduced to 676 articles. The title and abstract of each entry was read to identify and exclude non-epidemiology and non-related studies. Articles not written in English were also excluded. After this trimming, the count was reduced to 419 articles. Online and library searches for the full text of these 419 articles was conducted. Articles not available in the library of The Chinese University of Hong Kong were requested from other libraries in local universities. The total number of the articles for which the full text was retrieved was 331.

The full text of each of the 331 retrieved articles was read in order to determine inclusion or exclusion in the analysis. To be included in the analysis, the study must report epidemiology findings of injuries sustained in one or more sports activities, and should fulfill either one of the two following criteria: (i) the study must report injury at ankle joint, with either prevalence percentage compared with other body sites, or the incidence rate among the surveyed sample; (ii) the study must report ankle sprain injury, with either prevalence percentage among other ankle injury types, or the incidence rate among the surveyed sample. Review articles, case reports and current concepts were excluded. After the screening process, the final number of articles included in the analysis was 227.

Demographic data of the studies were summarised, which included the geographical location, period of the study, population, prospective or retrospective nature of the study and patient number. The geographical location was summarised in to the country and continent where the study was conducted. The period of study was categorised in 5-year intervals as pre-1980, 1981–5, 1986–90, 1991–5, 1996–2000 and 2001–5. Some investigations had a study period overlapping two or more 5-year inter-

vals. In such cases, the studies were counted in each of the relevant time intervals; therefore, the total number of counts exceeds the total number of studies. The populations investigated were split into the following five groups: (i) teams, where the subjects were from sports teams or clubs; (ii) public, where the subjects were from the public population attending accident and emergency departments in hospitals or sports medicine clinics; (iii) events, where the subjects were from competition events and games; (iv) schools, where the subjects were from schools and universities; and (v) military, where the subjects were military recruits. The number of patients were categorised as 1–100, 101–500, 501–1000, 1001–2000, 2001–5000, 5001–10 000 and >10 000.

The 227 included studies reported a total of 255 epidemiology reports, as some studies reported epidemiology in more than one sport. All epidemiology reports were categorised into the type of sport. Epidemiology studies reporting general sport injury patterns in a country, but not on specific sport, were categorised in to countries.

In each sport and country, the most common injured body sites and ankle injury types were ranked by weighted prevalence percentages. The following 11 body site groups were created: (i) ankle; (ii) foot; (iii) leg/shin/calf; (iv) knee; (v) thigh/hamstring; (vi) hip; (vii) trunk/back/spine; (viii) shoulder; (ix) arm/elbow; (x) hand/fingers; and (xi) head/neck/face. Some studies reported the percentage of injury of several body sites, such as foot and ankle, hip and thigh, or arm and shoulder. For these studies, the combined percentages were divided evenly for each included body site. Some studies combined the whole upper or lower extremity and made the distribution pattern among each body site too superficial. In these cases, the study was excluded as it failed to comply with the inclusive criteria that the study must report injury pattern in different body sites, with ankle injury reported. However, if an ankle and hand injury were separately reported from lower and upper extremities, the study was included and the percentages of lower and upper extremity were considered in the leg (leg/shin/calf) and arm (arm/elbow) categories. Ankle injury type was classified into the following 11 categories: (i) sprain; (ii) fracture; (iii) strain; (iv) abrasion; (v) contusion; (vi) tendinitis; (vii) blister; (viii) im-

pingement; (ix) cramp; (x) bruise; and (xi) laceration.

Among injured body sites, prevalence was represented by weighted percentages. This is calculated by the sum of the percentage of injury in a specific body site, times the number of all injuries in each included study, then divided by the sum of the total number of all injuries. Among ankle injury types, prevalence was represented by weighted percentages, calculated by the sum of the percentage of ankle injury type, times the number of all ankle injuries in each included study, then divided by the sum of the total number of all ankle injuries. The calculations are shown in the following mathematical formulae:

weighted percentage (body site) =

$$\frac{\sum_{i=1}^n \text{total case} \times \text{percentage (body site)}}{\sum_{i=1}^n \text{total case}}$$

weighted percentage (ankle injury type) =

$$\frac{\sum_{i=1}^n \text{total ankle injuries} \times \text{percentage (ankle injury type)}}{\sum_{i=1}^n \text{total ankle injuries}}$$

where n is number of epidemiology reports in one specific sport or country.

In addition, the incidence rate of ankle injuries and ankle sprains were reported in incidence per 1000 person-hour/person-year/person-season/person-exposure.

2. Results

2.1 Demographic Data

A total of 227 epidemiology studies were included. Some studies reported injury epidemiology in two or more countries and thus made the total count to 255. Most studies were conducted in Europe (45.5%), North America (30.2%) and Australasia (11.8%). Only a few studies were from Asia (4.3%), South America (2.0%) and Africa (0.4%). Distribu-

Table I. Geographical distribution of the studies included in this review

Continent	Countries
Europe (116)	UK (34), Sweden (18), Denmark (16), Holland (9), Finland (7), Germany (7), Norway (6), France (3), Greece (3), Iceland (2), Croatia (2), Czechoslovakia (1), Turkey (1), Switzerland (1), Austria (1), Belgium (1), Ireland (1), Italy (1), Spain (1), Poland (1)
North America (77)	US (62), Canada (11), Mexico (2), Guatemala (1), Dominica (1)
Australasia (30)	Australia (21), New Zealand (9)
Asia (11)	Hong Kong (4), Japan (2), Korea (1), Saudi Arabia (1), Singapore (1), Lebanon (1), Iran (1)
South America (5)	Argentina (2), Brazil (2), Trinidad and Tobago (1)
Africa (1)	Nigeria (1)
N/A (15)	
Total (255) ^a	

a The total count exceeds the number of studies as some studies reported epidemiology findings in more than one country.

N/A = data not available.

tion of geographical location of the studies is shown in table I. The results showed that most of the studies were conducted in the US (24.3%), followed by the UK (13.3%), Australia (8.2%), Sweden (7.1%) and Denmark (6.3%).

The period of study, population investigated, nature of the study and the patient number are summarised in table II. The majority of the studies were published between 1986 and 2000, with an even distribution in this time span. However, 37 studies (12.9%) did not report the duration of study. The populations investigated were predominantly from sports teams and clubs (40.5%), followed by the public (29.5%), competition events (15.4%), schools (11.0%) and the military (0.9%). Six studies (2.6%) did not describe the population investigated. The nature of study was identified to be prospective (75.3%) or retrospective (18.9%). Thirteen studies (5.7%) did not describe the study nature. Most studies (49.8%) had a patient number of 101–500. The total patient number of sports injuries was 201 600 and the number of ankle injuries included was 32 509.

2.2 Prevalence of Ankle Injuries in Different Sports

The total number of sports included in the analysis is 70. The weighted percentage of the most common injured body sites are shown in table III. Among these 70 sports, the ankle ranked the most popular injured body site in most studies (24 sports, 34.3%), followed by knee (14 sports, 20.0%), head (8 sports, 11.4%), trunk (6 sports, 8.6%) and hand (6 sports, 8.6%). Figure 1 shows the weighted percentage of ankle injury in all of the 70 included sports. Ankle injury was most common in aeroball (80.0%), wall climbing (60.0%), indoor volleyball (45.6%), mountaineering (40.0%), netball (39.8%), and field events in track and field (39.2%).

2.3 Prevalence of Ankle Sprains in Different Sports

Information about the ankle injury was available in 43 sports, and is shown in table IV. Ankle sprain was the most common injury (33 sports, 76.7%), followed by fracture (7 sports, 16.3%). Figure 2 shows the weighted percentage of ankle sprains in the 43 included sports. In some sports, all of the reported ankle injuries were ankle sprains (100%). These sports included Australian football, field hockey, handball, orienteering, scooter and squash. In addition to these six sports, ankle sprain percentage was also >80% in indoor volleyball, American football, martial arts, basketball, aeroball, ultimate frisbee, flag football, cheerleading, indoor soccer, ice hockey, lacrosse, badminton and netball.

2.4 Prevalence of Ankle Injury and Ankle Sprains in Different Countries

Epidemiology data in general sports were available in eight countries as shown in table V. In general, the knee was the most injured site (16–27%), followed by the ankle (11.2–20.8%), with an exception that the hand was most injured site in Sweden (19.8%) and Holland (21.8%). Information regarding the type of ankle injury was available in three countries (US, Holland and Hong Kong) and showed that sprains were the most common type of ankle injury (33.0–73.0%).

Table II. The period of study, population investigated, nature of study and patient number of the studies included in this review

Demographic	No. of studies
Period of study	
Pre-1980	17
1981–5	36
1986–90	53
1991–5	56
1996–2000	61
2001–5	26
N/A	37
Total	286^a
Population	
team	92
public	67
event	35
school	25
military	2
N/A	6
Total	227
Nature of study	
prospective	171
retrospective	43
N/A	13
Total	227
No. of patients	
1–100	52
101–500	113
501–1000	25
1001–2000	13
2001–5000	13
5001–10 000	10
>10 000	1
Total	227

a Total count exceeds the number of studies as some studies reported epidemiology findings in more than one period of time.

N/A = data not available.

2.5 Incidence of Ankle Injuries

The incidence of ankle injuries are shown in table VI. In terms of incidence per 1000 person-hour, hurling and camogie had the highest general incidence (32.88), followed by rugby (8.14), soccer (6.52), basketball (5.20) and triathlon (4.70). During games, the incidence was highest in soccer (34.83), followed by rugby (14.00), American football (13.80) and indoor soccer (11.68). In training, soccer showed the highest incidence rate (2.74). In

Table III. The most common injured body sites in different sports and countries^a

Sport	Most common injured body sites [weighted percentage (%)]					References
Adventure racing	Ankle (23.0)	Shoulder (11.5)	Arm (11.5)	Knee (11.0)	Trunk (11.0)	49
Aeroball	Ankle (80.0)	Knee (9.2)	Foot (6.4)	Hand (3.7)	Head (2.8)	50
American football	Knee (21.0)	Ankle (17.0)	Shoulder (8.4)	Trunk (5.8)	Thigh (4.9)	51-61
Australian football	Thigh (19.6)	Knee (12.3)	Ankle (7.9)	Leg (4.5)	Head (3.8)	36,62-65
Badminton	Ankle (23.5)	Knee (14.0)	Foot (12.5)	Arm (8.0)	Leg (5.4)	66-69
Baseball	Head (41.0)	Hand (29.0)	Leg (8.0)	Arm (6.0)	Ankle (3.5)	54
Basketball	Ankle (15.9)	Knee (10.7)	Trunk (6.5)	Thigh (5.4)	Leg (5.0)	23,54,57,70-82
Cardio kickboxing	Trunk (20.0)	Knee (18.0)	Hip (11.0)	Shoulder (10.0)	Thigh (8.0)	83
Cheerleading	Ankle (26.2)	Knee (10.4)	Trunk (9.4)	Arm (7.0)	Head (6.9)	26,84,85
Cricket	Ankle (17.9)	Knee (15.8)	Shoulder (14.7)	Trunk (12.6)		86
Cycling	Knee (24.3)	Hand (17.7)	Leg (11.9)	Foot (11.8)		57,71,87,88
Dancing	Leg (19.5)	Ankle (17.4)	Trunk (13.9)	Foot (12.0)	Knee (10.4)	12,15,89-94
Equestrian	Arm (22.3)	Head (21.7)	Hand (15.6)	Thigh (8.8)	Leg (7.0)	81,95
Fell walking	Head (28.8)	Ankle (24.0)	Leg (13.6)	Hand (9.1)	Trunk (7.2)	96
Field hockey	Head (34.4)	Ankle (34.0)	Arm (13.6)	Knee (4.1)	Trunk (1.1)	38
Figure skating	Ankle (24.5)	Leg (18.0)	Knee (15.6)	Head (9.8)	Arm (9.2)	35,97
Flag football	Hand (39.0)	Knee (18.0)	Thigh (9.0)	Ankle (8.0)	Foot (4.0)	98
Floorball	Ankle (26.8)	Knee (17.9)	Trunk (8.3)	Head (7.8)	Foot (5.1)	30,99
Gaelic football	Ankle (21.0)	Knee (13.0)	Thigh (13.0)	Shoulder (12.0)	Trunk (7.5)	100
Golf	Trunk (25.6)	Arm (21.2)	Shoulder (15.8)	Hand (14.9)	Head (6.1)	101,102
Gymnastics	Ankle (32.3)	Arm (25.9)	Knee (9.4)	Trunk (6.8)	Hand (4.4)	14,27,81,103-107
Handball	Hand (31.2)	Ankle (13.5)	Knee (10.7)	Arm (6.9)	Leg (6.6)	23,39,81,108,109
Hurling/camogie	Hand (27.1)	Head (20.2)	Ankle (8.6)	Leg (8.4)	Head (6.5)	110,111
Ice hockey	Head (21.4)	Knee (14.7)	Thigh (10.6)	Trunk (6.1)	Shoulder (5.6)	81,112-116
Jockey	Head (18.8)	Ankle (10.7)	Trunk (10.7)	Shoulder (9.6)	Knee (7.8)	117
Kitesurfing	Ankle (14.1)	Foot (14.1)	Head (13.7)	Trunk (12.9)	Knee (12.9)	118
Lacrosse	Knee (20.3)	Ankle (18.5)	Head (12.7)	Hand (9.9)	Shoulder (3.8)	119
Luging	Head (12.8)	Hand (11.3)	Trunk (8.9)	Shoulder (7.6)	Knee (7.6)	120
Marathon	Knee (31.7)	Foot (27.5)	Ankle (10.1)	Leg (6.8)	Thigh (1.3)	121
Martial arts	Arm (45.0)	Ankle (21.0)	Head (20.0)	Trunk (5.0)	Knee (4.4)	106,122
Motorcycle racing	Hand (22.4)	Arm (18.5)	Ankle (16.4)	Shoulder (14.6)	Knee (12.2)	123,124
Mountaineering	Ankle (40.0)	Knee (10.7)	Foot (7.9)	Leg (7.4)	Arm (5.6)	22
Netball	Ankle (39.8)	Knee (17.8)	Hand (10.5)	Head (8.7)	Leg (5.6)	1,30,76,125-127
Orienteering	Ankle (29.8)	Knee (18.1)	Leg (14.8)	Foot (13.3)	Thigh (5.0)	128-131
Parachuting	Ankle (32.9)	Leg (26.9)	Trunk (10.8)	Arm (3.1)	Hand (3.1)	132,133
Paragliding	Trunk (34.9)	Ankle (21.0)	Arm (13.4)	Knee (2.8)		134

Continued next page

Table III. Contd

Sport	Most common injured body sites [weighted percentage (%)]					References
Rodeo	Knee (16.9)	Trunk (12.4)	Head (11.6)	Shoulder (9.3)	Ankle (5.1)	135
Roller skating	Arm (32.8)	Hand (30.6)	Ankle (7.5)	Head (3.7)	Shoulder (3.3)	81,136-139
Rollerblading/inline skating	Hand (30.7)	Arm (16.4)	Knee (12.8)	Head (10.4)	Ankle (3.6)	54,140-142
Rugby	Head (13.7)	Thigh (12.7)	Ankle (11.6)	Knee (9.7)	Leg (8.4)	37,61,143-149
Running	Knee (28.9)	Ankle (20.2)	Foot (10.5)	Leg (7.6)	Thigh (4.7)	72,73,87,150,151
Scooter	Ankle (30.5)	Arm (25.6)	Hand (20.7)	Knee (4.9)	Head (2.5)	33,152
Skateboarding	Arm (24.5)	Hand (19.8)	Head (14.2)	Ankle (13.1)	Thigh (5.0)	81,153
Skiing (alpine)	Knee (32.2)	Head (11.0)	Ankle (7.6)	Leg (7.1)	Shoulder (6.5)	13,154-161
Skiing (nordic/telemark)	Knee (25.0)	Hand (13.4)	Shoulder (11.8)	Ankle (10.4)	Head (6.2)	158,162,163
Snowboarding/snowblading	Hand (17.9)	Ankle (15.6)	Knee (13.9)	Head (11.0)	Shoulder (8.9)	154,156,157,164-166
Soccer	Ankle (21.2)	Knee (16.3)	Thigh (12.2)	Leg (9.0)	Hip (4.3)	23,25,34,54,57,71,167,168 72,73,81,106,108,169-193
Soccer (indoor)	Ankle (23.7)	Knee (22.0)	Head (13.2)	Leg (7.4)	Foot (6.3)	194,195
Softball	Ankle (18.3)	Knee (11.1)	Hand (6.6)	Leg (4.4)	Head (2.2)	28,29,196
Speed skating	Knee (23.4)	Ankle (14.0)	Trunk (13.5)	Leg (12.6)	Hip (11.0)	197
Squash	Knee (20.8)	Trunk (15.9)	Ankle (12.6)	Shoulder (7.1)	Leg (2.5)	31,66
Swimming	Leg (34.9)	Foot (34.9)	Ankle (8.8)	Shoulder (6.9)	Hand (5.9)	71,87
Taekwondo	Leg (35.0)	Trunk (11.0)	Head (10.0)	Shoulder (10.0)	Ankle (8.0)	198
Tennis	Trunk (16.1)	Shoulder (11.0)	Thigh (8.6)	Ankle (8.0)	Foot (7.0)	66,199,200
Touch	Ankle (22.6)	Leg (16.4)	Knee (14.7)	Trunk (13.0)	Thigh (8.5)	201
Track and field (general)	Knee (21.8)	Leg (18.4)	Foot (13.0)	Ankle (11.2)	Hand (9.9)	71,150,202
Track and field (track)	Leg (28.1)	Thigh (18.9)	Trunk (18.7)	Ankle (9.4)	Foot (9.4)	150
Track and field (field)	Ankle (39.2)	Thigh (26.1)	Trunk (17.4)	Knee (17.4)	Leg (8.7)	150
Trampoline	Arm (48.1)	Head (6.3)	Ankle (6.2)	Leg (5.3)		203,204
Triathlon	Knee (22.5)	Trunk (11.8)	Foot (8.7)	Shoulder (8.2)	Ankle (7.8)	87,205,206
Tug of war	Trunk (42.0)	Knee (17.0)	Arm (12.0)	Shoulder (11.0)	Ankle (3.5)	207
Ultimate frisbee	Thigh (21.2)	Knee (15.1)	Ankle (12.2)	Leg (6.2)	Foot (6.2)	208
Ultramarathon	Ankle (30.9)	Knee (27.0)	Leg (13.0)	Thigh (11.0)	Foot (4.0)	209,210
Volleyball (indoor)	Ankle (45.6)	Arm (15.4)	Knee (11.4)	Shoulder (4.8)	Hand (4.3)	24,72,73,106,211-215
Volleyball (beach)	Knee (18.6)	Trunk (16.4)	Shoulder (14.6)	Ankle (12.4)	Hand (8.0)	213,216,217
Wakeboarding	Head (57.2)	Trunk (14.4)	Ankle (8.2)	Foot (6.4)	Knee (5.3)	218
Wall climbing	Ankle (60.0)	Shoulder (10.9)	Arm (7.3)	Hand (5.5)	Trunk (5.5)	219
Water skiing	Trunk (27.0)	Head (25.1)	Arm (13.0)	Knee (10.5)	Thigh (8.7)	218
Windsurfing	Foot (17.7)	Knee (9.4)	Trunk (8.9)	Ankle (8.6)	Head (7.4)	220
Wrestling	Knee (21.0)	Shoulder (13.9)	Head (11.0)	Ankle (9.0)		73,221

a Items with weight prevalence percentage <1% are omitted.

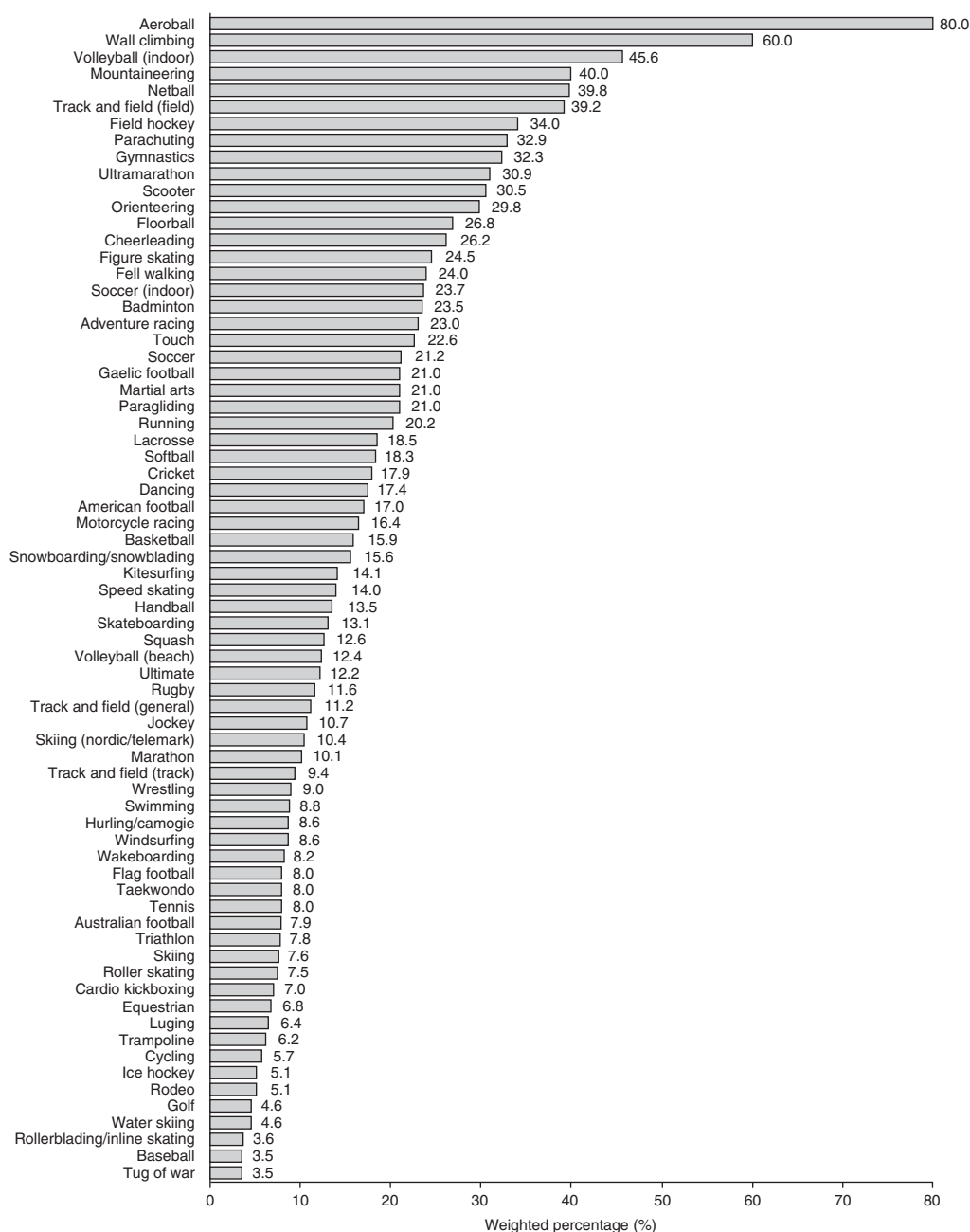


Fig. 1. Weighted percentage of ankle injury in the 70 different sports.

terms of incidence per 1000 person-year, field hockey showed the highest incidence rate (1000.00), followed by orienteering (571.43), dancing

(466.20), gymnastics (425.90) and gaelic football (374.70). When considering incidence per 1000 person-season, soccer showed the highest incidence in

Table IV. The most common ankle-injury type in different sports and countries^a

Sport	Most common ankle-injury type [weighted percentage (%)]					References
Aeroball	Sprain (90.0)	Fracture (10.0)				50
American football	Sprain (94.4)	Fracture (1.3)				51,56,61
Australian football	Sprain (100.0)					65
Badminton	Sprain (86.5)	Fracture (8.6)	Strain (5.4)			67,68
Basketball	Sprain (91.0)	Fracture (2.1)	Abrasion (0.7)			23,71,74,78,80,82,222
Cheerleading	Sprain (88.0)					85
Cycling	Abrasion (45.5)	Contusion (36.4)	Sprain (9.1)	Cramp (9.0)		71
Dancing	Sprain (50.5)	Fracture (5.6)				89,91
Field hockey	Sprain (100.0)					38
Figure skating	Sprain (77.0)	Fracture (23.5)				97
Flag football	Sprain (88.9)	Strain (11.1)				98
Gymnastics	Impingement (55.6)	Sprain (44.4)				107
Handball	Sprain (100.0)					23,39
Ice hockey	Sprain (87.7)	Fracture (8.3)				113,114,116
Kitesurfing	Sprain (77.8)	Fracture (22.2)				118
Lacrosse	Sprain (87.5)					119
Luging	Sprain (35.0)	Fracture (22.2)				120
Martial arts	Sprain (91.7)	Fracture (8.3)				122
Motorcycling	Sprain (31.3)					123
Mountaineering	Fracture (65.0)	Sprain (35.0)				22
Netball	Sprain (85.9)	Fracture (7.9)				125,223
Orienteering	Sprain (100.0)					32,128,129
Parachuting	Fracture (100.0)					132
Roller skating	Fracture (86.5)	Sprain (13.5)				136,137
Rollerblading/inline skating	Fracture (83.3)	Sprain (16.7)				141
Rugby	Sprain (74.9)					37,61,146,147
Scooter	Sprain (100.0)					33
Skiing	Sprain (76.9)	Fracture (11.0)				13,159
Skiing (nordic/telemark)	Sprain (69.0)	Fracture (31.0)				163
Snowboarding	Sprain (53.6)	Fracture (42.5)				164,166,224
Soccer	Sprain (76.8)	Bruise (2.8)	Tendinitis (2.2)	Fracture (1.1)		23,34,71,167,172,175,182
						184,186,187,189-192,225,226
Soccer (indoor)	Sprain (87.8)					194,195

Continued next page

Table IV. Contd

Sport	Most common ankle-injury type [weighted percentage (%)]	References
Softball	Fracture (74.6) Sprain (25.1)	28,29,196
Speed skating	Sprain (50.0)	197
Squash	Sprain (100.0)	31
Swimming	Cramp (66.7) Abrasion (33.3)	71
Tennis	Sprain (66.7)	200
Track and field (general)	Sprain (48.9) Abrasion (17.0) Cramp (17.0) Strain (17.0)	71,202
Trampoline	Fracture (100.0)	203
Ultimate frisbee	Sprain (89.9) Bruise (3.4) Strain (3.4)	208
Volleyball (indoor)	Sprain (99.3) Fracture (0.7)	Laceration (1.7)
Wall climbing	Fracture (57.6) Sprain (42.4)	24,215
Windsurfing	Sprain (58.6) Fracture (14.0)	219
		220

a Items with weight prevalence percentage <1% are omitted.

general (1200.00) games (143.40) and training (64.00). Australian football also has comparable game incidence rates (111.10). In terms of incidence per 1000 person-exposure, tennis showed the highest incidence (11.30), followed by basketball (9.17) and netball (5.26). During games, netball showed the highest incidence (45.60), followed by rugby (8.88), American football (6.38), basketball (3.77) and wrestling (2.71).

2.6 Incidence of Ankle Sprains

The incidence of ankle sprains is shown in table VII. In terms of incidence per 1000 person-hour, rugby had the highest general incidence (4.20), followed by soccer (2.52), volleyball (1.99), handball (1.59) and basketball (1.00). During games, the incidence was highest in soccer (11.68), followed by Australian football (4.86) and soccer (4.59). In terms of incidence per 1000 person-year, field hockey showed the highest incidence rate (1000.00), followed by rugby (233.40), basketball (173.50), dance (155.40) and American football (60.60). In terms of incidence per 1000 person-season, soccer showed the highest incidence in general (1200.00), while Australian football showed the highest incidence in game (111.10). When considering incidence per 1000 person-exposure, lacrosse showed the highest incidence (2.56), followed by basketball (1.90), alpine skiing (1.60), soccer (1.15) and field hockey (0.90). During games, netball showed the highest incidence (40.06), followed by rugby (6.31), American football (6.06) and basketball (3.45).

3. Discussion

This is a large-scale systematic review, which involved 227 published studies from 38 different countries. Injury patterns were reported in 70 different sports, with a total of 201 600 patients, including 32 509 patients with ankle injuries. A total of 14 098 ankle injuries were analyzed for the most common ankle-injury type, and 11 847 ankle sprains were included in this study. The review considered articles published over 29 years, from 1977 to 2005, inclusive. With such large coverage on the published data, contributed from researchers all around the world, this literature research should attract attention from those wishing in identifying the prob-

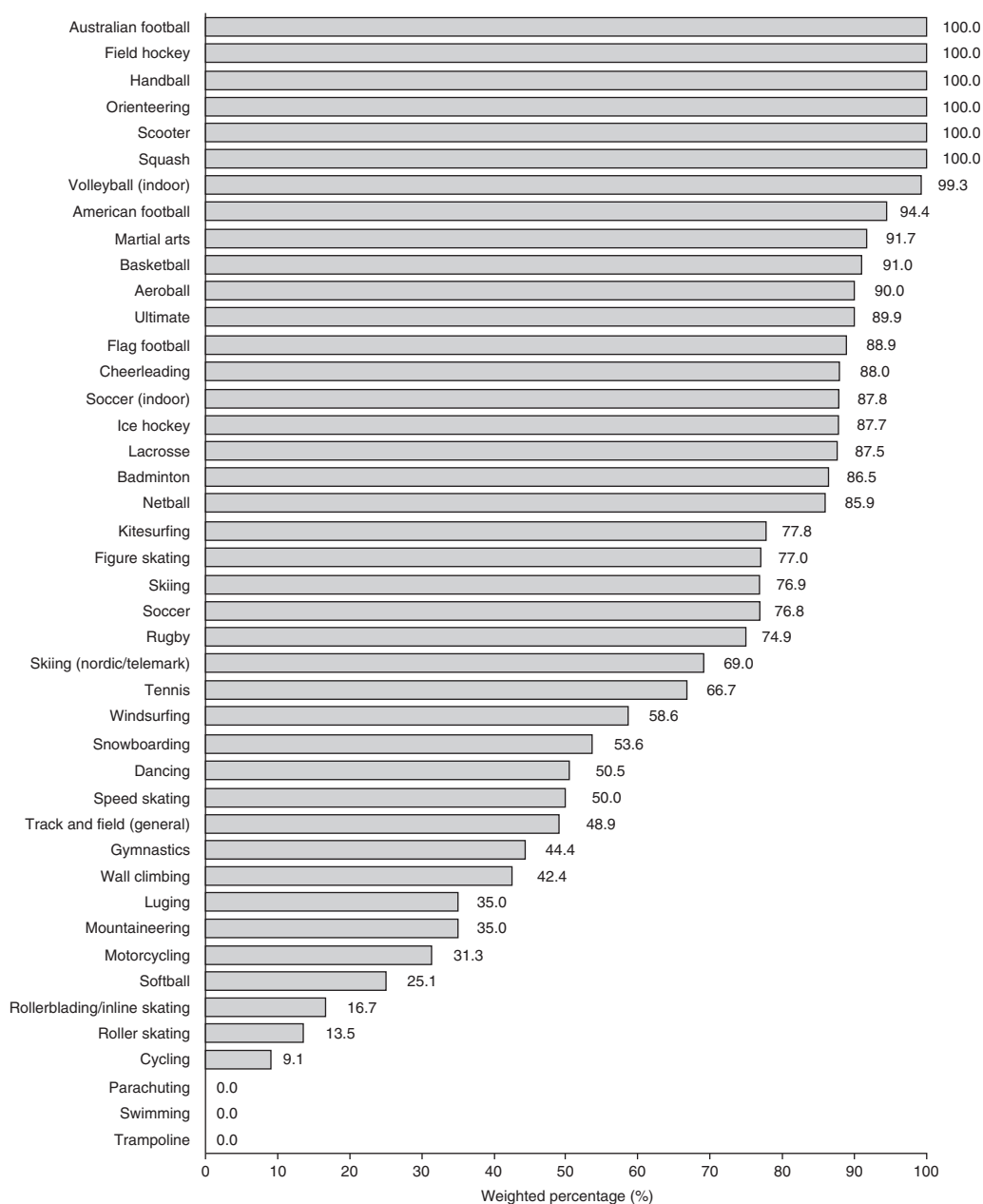


Fig. 2. Weighted percentage of ankle sprains in the 43 different sports.

lem of sports injury in injury prevention research. Although the details of injury in each sport cannot be listed out in this article, researchers may learn from the overview of published epidemiology data and be able to identify relevant articles for further

information. However, it is important to note that only those epidemiology studies in which ankle injury was reported were included in this analysis. This study can only provide information on ankle-

Table V. The most common injured body sites and type of ankle injuries in different countries

Country	Weighted percentage (%)			References		
Most common injured body site						
Canada	Knee (20.9)	Ankle (14.0)	Head (10.0)	Trunk (8.5)	Leg (7.5)	227
Finland	Knee (24.3)	Ankle (12.5)	Leg (9.0)	Trunk (6.8)	Shoulder (3.3)	228-230
Holland	Hand (21.8)	Ankle (20.8)	Foot (13.4)	Knee (13.2)	Thigh (4.1)	231,232
Hong Kong	Knee (27.0)	Ankle (11.2)	Foot (10.0)	Leg (10.0)	Hand (9.1)	71,72,233
Saudi Arabia	Knee (27.0)	Ankle (12.0)	Arm (11.0)	Thigh (10.0)	Hand (10.0)	234
Sweden	Hand (19.8)	Knee (16.0)	Ankle (15.6)	Foot (15.6)	Leg (6.6)	1
UK	Knee (18.3)	Ankle (15.1)	Hand (8.6)	Thigh (6.2)	Trunk (6.2)	235-239
US	Knee (22.1)	Ankle (12.3)	Leg (7.0)	Shoulder (5.8)	Trunk (4.9)	74,240-244
Most common ankle-injury type						
Holland	Sprain (73.0)	Fracture (16.0)	Contusion (5.0)			232
Hong Kong	Sprain (33.0)	Abrasion (25.2)	Cramp (17.4)	Contusion (13.0)	Strain (11.3)	71
US	Sprain (68.3)	Fracture (7.4)				12,40,74,240,241

injury patterns in sports, and not on general injury patterns.

Most epidemiology studies included in this analysis were from English speaking countries in North America, Europe and Australasia. There were fewer studies from Asia, South America and Africa. Literature searches showed that epidemiology studies had been published from these countries in their own languages; however, these were not included in this review because of the language barrier. Moreover, as a result of higher education level, scientific research activities were more active in those developed countries in North America, Europe and Australasia. Therefore, most of the included studies were contributed by these countries and thus results are more representative of the injury pattern in these countries rather than worldwide. This limitation is as a result of the language barrier and level of scientific research activities.

Some studies, especially those published some time ago, did not fully describe the demographic details of the study. The information was obvious, but sometimes the authors just did not report them. For example, some authors did not report the location in which the study was conducted.^[11,12,15,51-53,70,89,167,168,211,216] Others only reported the city or town name and sometimes it was impossible for international reader to identify the exact location.^[16,54,202] Information on the nature of study, the period of data collection and the participants investigated were also often omitted, as seen in table II. Although these data might not always be important to every reader, reporting this information enriches the completeness of the published epidemiology study.

There were very few published epidemiology studies reporting sports with ankle injury before 1980. After 1981, the number of these studies increased and remained at a steady level until 2000. From 2001 to 2005, the number of recorded studies dropped. This may reflect a decline of interest in epidemiology studies, a shift in the sports events of interest (involving fewer ankle injuries) or maybe a total vanish of ankle injuries in some sports in recent years. The populations investigated were mainly team athletes (40.5%) and public participants (29.5%), which is a good representation of organised and recreational sports participation. Three-

Table VI. The ankle-injury incidence rate in different sports (game incidence [G] and training incidence [T] in brackets)

Sport	Ankle-injury incidence rate			
	per 1000 person-hour	per 1000 person-year	per 1000 person-season	per 1000 person-exposure
Adventure racing	(G = 0.30 ^[49])			
Aeroball		1.37 ^[50]		
American football	0.54–4.15 ^[52,55,58] (G = 13.80 ^[59])	26.30–72.00 ^[51,56,60]		0.71 ^[245] (G = 1.23–6.38 ^[53,61])
Australian football	1.30–2.31 ^[62,64] (G = 3.91–4.86 ^[36])		(G = 111.10 ^[65])	
Badminton	0.68 ^[68]			
Baseball				0.14 ^[245]
Basketball	1.00–5.20 ^[23,75,77,79]	178.60 ^[222]		0.47–9.17 ^[70,78,245,246] (G = 3.62–3.77 ^[76,82])
Cheerleading	0.28–0.33 ^[26,84]			0.21–0.34 ^[85,245]
Cricket	1.21 ^[86]			
Dancing	0.32–2.66 ^[12,91,93]	117.10–466.20 ^[90,94]	131.40 ^[15]	
Equestrian		0.04–10.30 ^[95]		
Field hockey		1000.00 ^[38]		
Floorball	0.87 ^[30]			
Gaelic football		374.70 ^[100]		
Golf			20.60 ^[102]	
Gymnastics	0.12–0.62 ^[27,105] (T = 1.03 ^[104])	230.80–425.90 ^[103,107]	40.80–46.70 ^[14]	
Handball	0.38–1.59 ^[23,39,109] (G = 1.32 ^[108])			
Hurling/camogie	32.88 ^[111]			
Ice hockey	0.12–0.20 ^[113,114] (G = 1.53; ^[113] T = 0.18 ^[113])	(G = 64.81 ^[117])		0.46–0.71 ^[112,116]
Jockey				
Kitesurfing	2.0 ^[118]			
Lacrosse				0.47–2.66 ^[119]
Luging				0.45 ^[120]
Motorcycle racing	3.73 ^[124] (G = 2.85–6.15; ^[124] T = 0.00 ^[124])	(G = 4.50 ^[123])		
Netball	3.27 ^[126]			5.26 ^[76] (G = 45.60 ^[223])

Continued next page

Table VI. Contd

Sport	Ankle-injury incidence rate			
	per 1000 person-hour	per 1000 person-year	per 1000 person-season	per 1000 person-exposure
Orienteering	0.82 ^[130] (G = 3.80 ^[32])	571.43 ^[131]		(G = 1.03 ^[129])
Parachuting		0.38 ^[133]		0.45 ^[132]
Rodeo				(G = 0.75 ^[135])
Rugby	0.23–8.14 ^[146,147] (G = 5.77–14.00; ^[144,149] T = 0.26 ^[143])	233.40 ^[37]		(G = 2.72–8.88 ^[61,145])
Running	0.02 ^[151]	62.50–111.00 ^[150]		
Skiing (alpine)				0.13–3.18 ^[13,158,159]
Skiing (nordic/telemark)				0.07 ^[158]
Snowboarding/snowblading				0.97 ^[154] (G = 0.13 ^[166])
Soccer	0.17–6.52 ^[23,167,170,173-177,179,181,183,188-193] (G = 0.00–34.83; ^[2,5,108,169,170,172,173,177,180,184,187,225] T = 0.21–2.74 ^[170,173])	71.13 ^[34]	10.99–1200.00 ^[25,178,182,186,226] (G = 39.79–143.40; ^[178,186] T = 36.35–64.00 ^[178,186])	0.55–0.76 ^[245]
Soccer (indoor)	(G = 11.49–11.68 ^[194,195])			
Softball		41.70 ^[196]		0.41 ^[245]
Speed skating		168.40 ^[197]		
Tennis				11.30 ^[199] (G = 0.30 ^[200])
Touch	1.10 ^[201]			
Track and field (general)		29.90–102.00 ^[150,202]		0.13–0.14 ^[245]
Track and field (track)		0.00–111.00 ^[150]		
Track and field (field)		158.00–286.00 ^[150]		
Trampoline		0.02–0.05 ^[203]		
Triathlon	1.49–4.70 ^[205]			
Ultimate frisbee		(G = 14.75 ^[208])		
Volleyball (indoor)	0.60–2.01 ^[24,212,213,215] (G = 5.50 ^[214])			0.68 ^[245]
Volleyball (beach)	0.20 ^[213] (G = 0.57; ^[217] T = 0.13 ^[217])			
Wall climbing				0.02 ^[219]
Windsurfing				0.09 ^[220]
Wrestling				0.09–0.86 ^[245,247] (G = 2.71; ^[247] T = 0.63 ^[247])

Table VII. The ankle-sprain incidence rate in different sports (game incidence [G] and training incidence [T] in brackets)

Sport	Ankle-sprain incidence rate			
	per 1000 person-hour	per 1000 person-year	per 1000 person-season	per 1000 person-exposure
Aeroball		1.23 ^[50]		
American football		59.40–60.60 ^[51,56]		(G = 6.06 ^[61])
Australian football	(G = 3.69–4.86 ^[36,248])		(G = 111.10 ^[65])	
Badminton	0.53 ^[68]			
Basketball	1.00 ^[23]	173.50 ^[222]		0.42–1.90 ^[78,249] (G = 3.45 ^[82])
Cheerleading				0.18 ^[85]
Cricket				
Dancing	0.08–0.30 ^[91,93]	155.40 ^[94]		
Field hockey		1000.00 ^[38]		0.90 ^[249]
Floorball				
Gymnastics		102.60 ^[107]		
Handball	0.38–1.59 ^[23,39]			
Ice hockey	0.05–0.12 ^[113,114]			0.35–0.62 ^[112,116]
Kitesurfing	0.4 ^[118]			
Lacrosse				0.39–2.56 ^[119,249]
Luging				0.16 ^[120]
Motorcycle racing		(G = 1.40 ^[123])		
Netball				(G = 40.06 ^[223])
Orienteering				(G = 1.03 ^[129])
Parachuting				0.00 ^[132]
Rugby	0.15–4.20 ^[146,147]	233.40 ^[37]		(G = 6.31 ^[61])
Skiing (alpine)				0.36–1.60 ^[13,159]
Snowboarding/snowblading				(G = 0.13 ^[166])
Soccer	0.15–2.52 ^[23,167,170,174,175,177,189–192,250] (G = 0.34–4.59 ^[170,184,187,225,251] T = 0.21 ^[170])	71.13 ^[34]	54.79–1200.00 ^[182,186,226]	0.73–1.15 ^[249]
Soccer (indoor)	(G = 9.65–11.68 ^[194,195])			
Softball		41.7 ^[196]		
Speed skating		84.20 ^[197]		
Tennis				(G = 0.20 ^[200])
Track and field (general)		29.90 ^[202]		
Trampoline		0.00 ^[203]		
Ultimate frisbee		(G = 13.30 ^[208])		
Volleyball (indoor)	1.37–1.99 ^[24,215]			
Volleyball (beach)				
Wall climbing				0.01 ^[219]
Windsurfing				0.05 ^[220]

quarters (75.3%) of the studies were prospective, which are expected to provide a more realistic representation of injury patterns as a result of better experimental control and data collection procedure. Most studies (49.8%) had 101–500 patients, which is sufficient for epidemiology data reporting. Therefore, we believe that the summarised findings are representative and should have significant value in helping identify the ankle injury patterns.

Within the 70 included sports, ankle injury was most common in 24 sports, with the highest weighted percentage in aeroball (80.0%), wall climbing (60.0%), indoor volleyball (45.6%), mountaineering (40.0%), netball (39.8%) and field events in track and field (39.2%). Within the 43 sports providing information about ankle injury, ankle sprain was the most common injury in 33 sports. The weighted percentage of sprain among all ankle injuries reached 100% in 6 sports, >80% in 19 sports, >60% in 26 sports and >40% in 34 sports. Among the eight countries included in this study, the knee was the most injured site (16.0–27.0%), followed by the ankle (11.2–20.8%). Exceptions to this were seen in Sweden, where the hand was most common (19.8%), followed by the knee (16.0%) and then the ankle (15.6%); and in Holland, where the hand was the most common (21.8%), followed by the ankle (20.8%) and then the foot (13.4%). Sprain was the most common type of ankle injury (33.0–73.0%) from the information from the US, Holland and Hong Kong. From these results, ankle injury was concluded to be one of the most commonly injured body sites in sports and ankle sprain was the most dominating type of ankle injury in the majority of the sports.

In studying the weighted percentage of the injured body sites, it should be noted that a higher percentage did not necessarily mean a higher incidence of injury to this site. The figures only represent the distribution of the recorded injury in different sites. For example, an 80.0% ankle injury rate in aeroball does not mean a higher injury occurrence than a 21.2% ankle injury rate in soccer, as the incidence rate of injury in soccer (71.13 incidence per 1000 person-year) was much higher than that in aeroball (1.37 incidence per 1000 person-year). From this review, the incidence of ankle injuries and ankle sprains was high in court games and team

sports, such as rugby, soccer, volleyball, handball, basketball, American football, Australian football, Gaelic football, tennis, netball, lacrosse and field hockey. Moreover, the incidence of ankle injury was also high in gymnastics, orienteering, dancing and skiing. In general, the incidence of ankle injury in competitive games was higher than in other sports.

It is important to note that a higher incidence rate does not necessarily mean higher severity of the sports injury. It only reports the frequency of injury occurrence and does not report the severity of each injury. In ankle sprains, the severity can be ranked in grades (e.g. grade 1, 2 and 3). However, several different rating scales for ankle sprains were used. Moreover, such information was rarely reported in the epidemiology studies and thus was not summarised in this review.

A high incidence rate does not give all the information about the sports injury. For example, among the listed sports with high ankle-injury and ankle-sprain incidence, soccer may have the highest number of ankle sprains as it is a very popular sport throughout world. Although lacrosse showed a higher ankle-sprain incidence (2.56 incidence per 1000 player-exposure) than soccer (1.15 incidence per 1000 player-exposure), the total number of ankle sprains sustained in lacrosse may be fewer because the sport is not as popular. Information on the total amount of incidence was not available from this review. Garrick and Requa^[16] conducted a prospective epidemiology study in the US from 1979 to 1987, with 19 sports and 17 750 patients. Among these patients, most of the injuries were sustained during running (≈ 4200 , 24%). The total number of ankle injuries were highest in running (≈ 350), followed by ballet (≈ 220), basketball (≈ 130), tennis (≈ 90) and soccer (≈ 70). Although the percentages of ankle injuries were higher in soccer and basketball in the study by Garrick and Requa,^[16] the total patient number was fewer than that of running as a result of smaller participation rate. From the results of this study, the incidence rate per 1000 person-hour was 0.02 for running, 5.20 for basketball and 6.52 for soccer. This showed that we should also consider the popularity of a specific sport, or the total participants in this specific sport, in order to decide which sport we should select to implement injury prevention measures.

The results from this study may help sports medicine specialists in deciding which sports to target when implementing injury prevention protocols to prevent ankle sprain. It can also serve as a standard or baseline for sports teams to evaluate their injury pattern by comparing it with the reported data in this review. If a team records a higher ankle-injury or ankle-sprain incidence than those found in the literature, it may be necessary to closely examine the injuries, sports equipment and facilities. Moreover, when injury prevention protocols were introduced to a certain population, sports medicine specialists can evaluate their effectiveness by investigating the injury patterns and incidence with the data reported in this study.

4. Conclusion

Ankle injury was most common in aeroball, wall climbing, indoor volleyball, mountaineering, netball and field events in track and field. Among ankle injuries, ankle sprains were the most common type of injury. In Australian football, field hockey, handball, orienteering, scooter and squash, all of the reported ankle injuries were sprains. Sprains were also very common in other sports, especially court games and team sports. However, in the epidemiology of general sports injuries, knee injuries were the most common, preceding ankle injuries. In the US, Holland and Hong Kong, ankle sprains were the most common ankle injury type. Ankle-injury and ankle-sprain incidences were highest in court games and team sports, such as rugby, soccer, volleyball, handball and basketball.

This systematic review provides a summary of the epidemiology of ankle injury in sports. It helps sports medicine specialists when deciding which sport to implement injury prevention measures. However, one should also pay attention to the popularity of sports, as a high participation rate would also significantly increase the total number of patients with injuries. We recommend that ankle-sprain prevention should be implemented in rugby, soccer, volleyball, handball and basketball. Ankle-sprain prevention would also be beneficial in running, as it is the most popular sport in the world with the most participants.

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Correspondence and offprints: Professor *Kai-Ming Chan*, Department of Orthopaedics and Traumatology, Prince of Wales Hospital, Faculty of Medicine, The Chinese University of Hong Kong, Shatin, Hong Kong, China.
E-mail: kaimingchan@cuhk.edu.hk